

Chapter-4:

Operating System Basics

Syeda Jannatul Naim
Lecturer | Department of CSE

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1. What Is an Operating System?

❖ **The Core Software**

The master program that manages all hardware and software resources. It starts when we power on (boot) our computing device and runs continuously.

❖ **The Bridge**

Acts as an intermediary between us (the user) or our applications and the complex computer hardware. We don't speak binary; the OS does.

❖ **Provider of Services**

Handles essential background tasks like file saving, network connectivity, printing documents, and enforcing security.

1.2. Popular Operating System



DESKTOP

Microsoft Windows

The standard for pharmacy workstations. High compatibility with legacy dispensing hardware.

Pharmacy Use:

Main dispensing computers, insurance billing, patient record management (EHR).



DESKTOP

macOS

Known for stability and security. Less common in dispensing but popular in admin.

Pharmacy Use:

Hospital administration, medical research, educational tools.



SERVER

Linux

Open-source, secure, and stable. Runs the "back-end" infrastructure invisibly.

Pharmacy Use:

Database servers for drug interactions, MRI/CT scanners, embedded devices.



MOBILE

Android

Versatile mobile OS running on various hardware from phones to scanners.

Pharmacy Use:

Handheld inventory scanners, delivery driver tablets, point-of-sale systems.



MOBILE

iOS / iPadOS

High security and consistent interface. Very popular for clinical mobility.

Pharmacy Use:

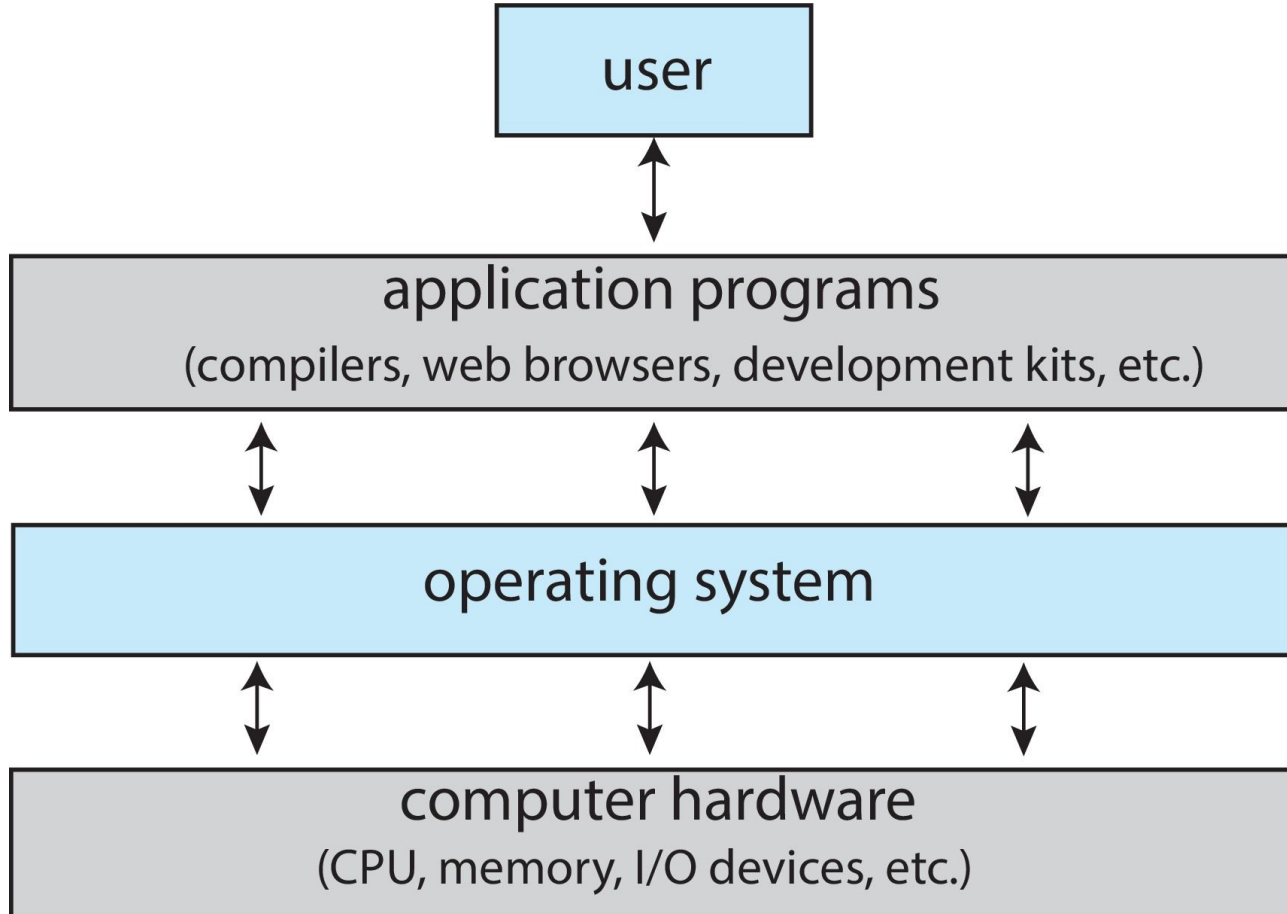
Bedside charting (eMAR), secure messaging apps, drug reference apps.

Interconnection

Modern pharmacy relies on all of these together.

Example: A pharmacy could use workstations on Windows, servers on Linux, and handheld devices on Android or iOS. All of these systems would be interconnected.

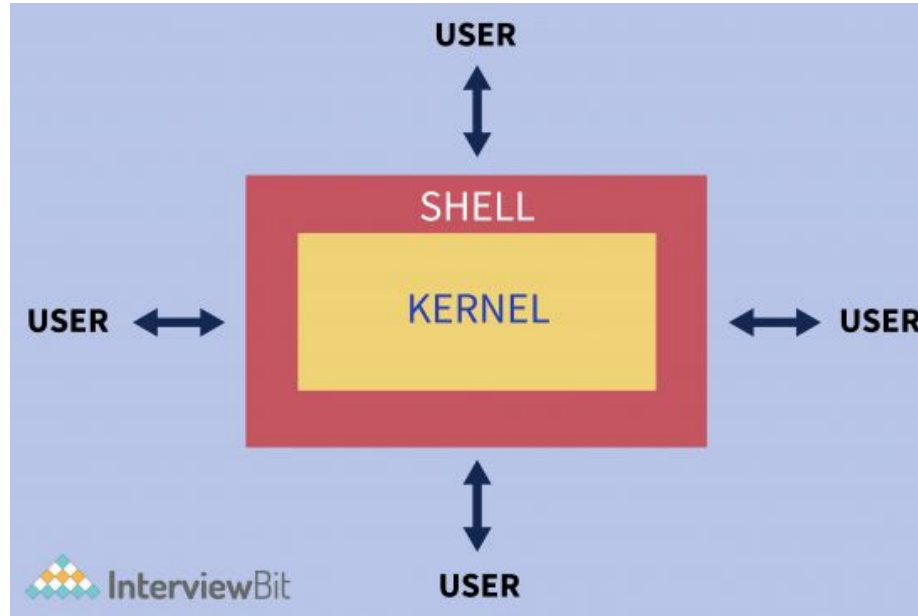
2. Abstract view of Operating System:



3. Components of Operating System

The operating system consists of two components:

1. Shell
2. Kernel



3. Components of Operating System

Kernel: A kernel is an operating system's heart (Operating system). Its job is to keep track of the hardware and computer's operations. By accessing the different resources of a computer, such as the I/O devices, CPU, and several other resources, a kernel simply works as a bridge between any user and the various resources supplied by a system.

Shell: It refers to a specific user software or environment that gives a user an interface to access the operating system's services. A shell runs several programmes based on the input it receives from the user. Shell commands such as mkdir, ls, and others can be used to ask the operating system to complete a specified action (OS).

4.Key functions(services) of an operating system



Process Management

Creates, schedules, and terminates running programs (processes).



Memory Management

Allocates RAM to apps and prevents them from interfering with each other.



File System

Organizes data into files/folders and manages storage space.



Device (I/O) Mgmt

Communicates with hardware like printers, scanners, and keyboards.



CPU Scheduling

Decides which task gets to use the processor and for how long.



Security

Manages user logins, permissions, and protects against unauthorized access.



Networking

Handles data transmission over Wi-Fi, Ethernet, and the Internet.



Error Handling

Detects system crashes or hardware faults and logs them.

5. Importance of Operating System

- An operating system is essential for **starting and running a computer**.
- It acts as an **interface between the user and computer hardware**.
- It allows users to **run application software** like Word, browsers, and pharmacy software.
- It **manages computer resources** such as CPU, memory, and storage.
- It helps in **file and data management**, including storing and retrieving information.
- It supports **multitasking**, allowing multiple programs to run at the same time.
- It provides **security and data protection** through passwords and access control.
- It controls **input and output devices** like keyboard, printer, scanner, and medical devices.

